

Topic 02: What is a Matrix, Really? Linear Transformations

Applied Linear Algebra via NumPy

TOPIC 02

What is a Matrix, Really? Linear Transformations

Matrices as spatial deformation operators, basis shifts, determinants, and rank.

Cognitive Objectives

By the end of this topic, learners will be able to:

- **Visualize matrices as spatial operators** that stretch, rotate, and shear coordinate space rather than static number grids.
- **Track basis transformations** by mapping where standard unit vectors $\hat{i}$ and $\hat{j}$ land.
- **Interpret determinants geometrically** as area/volume scaling factors ($\det(\mathbf{A}) = 0 \implies$ irreversible compression).
- **Diagnose linear systems** using Column Space $\mathcal{C}(\mathbf{A})$, Null Space $\mathcal{N}(\mathbf{A})$, and the Rank-Nullity Theorem.

The Disappearing Dimension Mystery

The Collision: Matrix as Compression

Consider $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

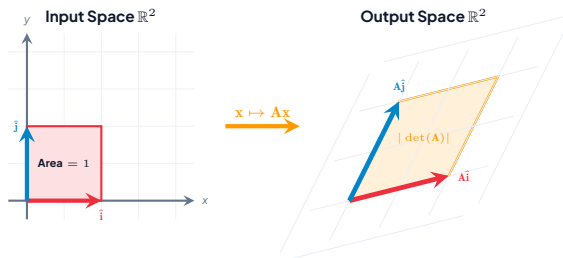
- Multiply 10,000 diverse 2D data points $x \in \mathbb{R}^2$ by A .
- Every single transformed point Ax lands on the line $y = 2x$!
- **Socratic Question:** Can we ever reconstruct the original 2D points from the output?

The "Aha!" Discovery: Landing Sites

- Standard basis: $\hat{i} = [1, 0]^T$ and $\hat{j} = [0, 1]^T$.
- $A\hat{i} = [1, 2]^T$ (Col 1 of A).
- $A\hat{j} = [2, 4]^T = 2 \cdot [1, 2]^T$ (Col 2).
- Columns are linearly dependent ($c_2 = 2c_1$).
- The entire 2D universe is flattened into a 1D line!

PUNCHLINE Columns tell you where the standard basis vectors land.

Hero Visualization: Grid Deformation Under A

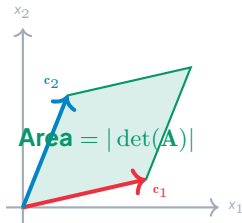


Matrices as Spatial Operators

- **Line of Action:** Every linear transformation keeps the origin fixed and keeps grid lines parallel and evenly spaced.
- **Coordinate Recipe:** Any vector $\mathbf{v} = x\hat{i} + y\hat{j}$ transforms to:
$$A\mathbf{v} = x(A\hat{i}) + y(A\hat{j})$$
- Matrix multiplication is simply taking linear combinations of the matrix columns!

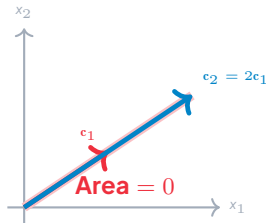
Hero Visualization: Determinants & Information Collapse

Non-Singular Matrix ($\det(A) \neq 0$)



Properties: Area is scaled by $|\det(A)|$. Full Rank (Rank = 2). Transformation is **reversible** (A^{-1} exists).

Singular Matrix ($\det(A) = 0$)



Properties: Area collapsed to zero. Rank deficient (Rank = 1). Null space is 1D line (Nullity = 1). **Irreversible** information loss.

Production Checkpoint: Solving Linear Systems in NumPy

The Anti-Pattern: Explicit Matrix Inversion

```
1 # DO NOT DO THIS IN PRODUCTION:
2 A_inv = np.linalg.inv(A) # O(n^3) ops
3 x = A_inv @ b           # Severe roundoff errors!
```

Flaw: Explicit inversion is $3 \times$ slower and numerically unstable near ill-conditioned matrices ($\kappa(A) \gg 1$).

The Enterprise Pattern: LU / LAPACK Solve

```
1 # PRODUCTION-GRADE LAPACK SOLVER:
2 # Solves Ax = b via LU decomposition
3 x = np.linalg.solve(A, b)
4
5 # Verify condition number before solving
6 cond = np.linalg.cond(A)
7 if cond > 1e12:
8     raise ValueError(f"Ill-conditioned A: {cond}")
```

PUNCHLINE Always use `np.linalg.solve`
over `np.linalg.inv`.